Applied Econometrics: Session 6
Matching and Propensity Scores

Swapnil Singh

This lecture studies matching and propensity-score methods for observational causal inference. The main question is simple: when treatment is not randomly assigned, can we construct a credible comparison group using observed pre-treatment covariates? We cover conditional independence, overlap, ATE and ATT, exact matching, nearest-neighbor matching, common support, the propensity-score theorem, propensity-score matching, inverse probability weighting, diagnostics, and reporting standards.

> **Reading and objective.** Read Chapters 10 and 11 of Matheus Facure Alves, *Causal Inference for the Brave and True*: `https://matheusfacure.github.io/python-causality-handbook/10-Matching.html` and `https://matheusfacure.github.io/python-causality-handbook/11-Propensity-Score.html`. The companion script is `lectures/code/lecture-6-matching-propensity.py`. The objective is to learn how to design, diagnose, estimate, and report a selection-on-observables study without pretending that software has solved the identification problem.

The Observational Problem

We observe a sample

$$(Y_i, D_i, X_i), \qquad D_i \in \{0, 1\},$$

where Y_i is the observed outcome, D_i is the treatment indicator, and X_i is a vector of pre-treatment covariates. The potential outcomes are Y_{1i}, the outcome unit i would have under treatment, and Y_{0i}, the outcome unit i would have without treatment. The observed outcome is

$$Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}.$$

For each unit we observe only one potential outcome. If $D_i = 1$, then $Y_i = Y_{1i}$ and Y_{0i} is missing. If $D_i = 0$, then $Y_i = Y_{0i}$ and Y_{1i} is missing. This is the fundamental missing-data problem of causal inference.

In a randomized experiment, treated and untreated units are comparable in large samples because assignment is independent of potential outcomes. In an observational study, treatment is often chosen by people, firms, doctors, schools, or governments. Treated and untreated units can differ even before the treatment occurs. Those

pre-treatment differences may also predict outcomes. Then the raw treated-control difference,

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0],$$

mixes the causal effect with selection bias.

Definition 1 (Selection bias)
Selection bias is the difference between the observed comparison group and the missing counterfactual group. For the treated population,

$$\begin{aligned}&\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] \\ &\quad = \mathrm{ATT} + \{\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]\}.\end{aligned}$$

The second term is selection bias. It is zero in a randomized experiment, but not generally zero in observational data.

Example 1 (Training program)
Suppose participants in a job-training program earn more after the program than non-participants. Is that the effect of training? Maybe. But participants may also be younger, more motivated, closer to employment offices, or more likely to have recent employment histories. If those characteristics would have raised earnings even without training, the naive gap overstates the effect. Matching asks whether we can compare trained and untrained people with similar pre-treatment characteristics.

Identification Assumptions

Assumption 1 (Conditional independence)
The conditional independence assumption, also called selection on observables or unconfoundedness, is

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i \mid X_i.$$

After conditioning on X_i, treatment assignment is as good as random.

This assumption is not a mechanical assumption. It is a claim about the economic assignment process. If the treatment is a scholarship, X_i must contain the pre-treatment determinants of both scholarship receipt and later outcomes: prior grades, family background, school characteristics, location, and so on. If unobserved motivation affects both scholarship receipt and later income, conditioning on observed X_i will not remove that bias.

> **Important warning.** Matching and propensity scores do not fix selection on unobservables. They organize comparisons among units that look similar in observed variables. If the important

> difference is unobserved, these methods can still be badly biased.

Assumption 2 (Overlap)
For the covariate values relevant to the target estimand,

$$0 < \mathbb{P}(D_i = 1 \mid X_i = x) < 1.$$

There must be treated and untreated units with comparable X_i.

Overlap is the empirical content behind the phrase "comparable group." If all high-income urban workers receive training and no high-income urban worker is untreated, the data contain no direct evidence about what those treated workers would have earned without training. An estimator can extrapolate, but that is modeling, not matching.

ATE, ATT, and ATU

Three estimands appear often:

$$\text{ATE} = \mathbb{E}[Y_{1i} - Y_{0i}],$$

$$\text{ATT} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1],$$

$$\text{ATU} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 0].$$

The ATE averages over the whole population. The ATT averages over the treated population. The ATU averages over the untreated population. If treatment effects are heterogeneous, these are different objects.

Many matching exercises naturally estimate the ATT because we start with each treated unit and search for untreated matches to impute its missing Y_0. IPW can be written to target either the ATE or the ATT. Therefore the estimand must be chosen before estimation. Do not report "the effect" without saying whose effect.

Exact Matching

Exact matching is the cleanest version of the idea. Suppose X_i takes a small number of discrete values. For a covariate cell x, define the within cell treatment-control difference

$$\delta(x) = \mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x].$$

Under conditional independence and overlap,

$$\delta(x) = \mathbb{E}[Y_{1i} - Y_{0i} \mid X_i = x].$$

For the ATT,

$$\text{ATT} = \sum_x \delta(x)\mathbb{P}(X_i = x \mid D_i = 1).$$

The weights are the distribution of X_i among treated units, because the target population is the treated population.

Example 2 (A tiny exact-matching calculation)
Suppose four covariate cells produce the following cell effects and treated shares.

Cell	*Cell effect*	*Treated share*	*Contribution*
Young, urban	2.5	0.40	1.00
Young, rural	2.0	0.20	0.40
Older, urban	1.5	0.25	0.375
Older, rural	1.0	0.15	0.150

The exact-matching ATT is

$$1.00 + 0.40 + 0.375 + 0.150 = 1.925.$$

Notice that we do not compare all treated units with all controls. We compare within cells, then average using treated-population weights.

Exact Matching as the G-Formula

Start from

$$\text{ATT} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1].$$

Iterate over covariates:

$$\begin{aligned}\text{ATT} = \sum_x \{&\mathbb{E}[Y_{1i} \mid X_i = x, D_i = 1] \\ -&\mathbb{E}[Y_{0i} \mid X_i = x, D_i = 1]\}\, \mathbb{P}(X_i = x \mid D_i = 1).\end{aligned}$$

The first conditional mean is observed:

$$\mathbb{E}[Y_{1i} \mid X_i = x, D_i = 1] = \mathbb{E}[Y_i \mid X_i = x, D_i = 1].$$

For the missing mean, conditional independence gives

$$\begin{aligned}\mathbb{E}[Y_{0i} \mid X_i = x, D_i = 1] &= \mathbb{E}[Y_{0i} \mid X_i = x, D_i = 0] \\ &= \mathbb{E}[Y_i \mid X_i = x, D_i = 0].\end{aligned}$$

Substituting yields the exact-matching formula above. This derivation is useful because it shows precisely where the identifying assumption enters.

Matching as Imputation

Exact matching becomes difficult when X_i is continuous or high dimensional. Nearest-neighbor matching replaces exact equality with closeness. For a treated unit i, let $\mathcal{N}_K(i; D = 0)$ be the set of its K nearest untreated neighbors. Then

$$\widehat{Y}_{0i} = \frac{1}{K} \sum_{\ell \in \mathcal{N}_K(i;D=0)} Y_\ell,$$

and the matching estimator for the ATT is

$$\widehat{\text{ATT}}_M = \frac{1}{N_1} \sum_{i:D_i=1} (Y_i - \widehat{Y}_{0i}).$$

This is imputation. We fill in the missing untreated potential outcome for each treated unit using observed untreated outcomes from similar units.

Definition 2 (Distance metric)
A common distance between treated unit i and control unit ℓ is

$$d(i, \ell) = \left[(X_i - X_\ell)' W (X_i - X_\ell)\right]^{1/2}.$$

If $W = I$, this is Euclidean distance after scaling. If $W = \widehat{\Sigma}_X^{-1}$, this is Mahalanobis distance.

Scaling is not optional. If one variable is income in euros and another is age in years, raw Euclidean distance will usually be dominated by income. Standard practice is to standardize continuous covariates before distance matching.

Replacement, Calipers, and Neighbors

Matching with replacement allows the same control unit to be used many times. This often improves match quality because each treated unit can use the closest available control. The cost is that the estimator may depend heavily on a small number of controls.

Matching without replacement uses each control at most once. This can increase the number of distinct controls, but later treated units may receive poor matches because the best controls have already been used.

A caliper imposes a maximum acceptable distance:

$$d(i, \ell) \leq c.$$

Calipers prevent visibly bad matches. The cost is that some treated units are dropped, so the estimand may become the ATT for the matched treated population, not the ATT for all treated units.

Bias-variance tradeoff. One close neighbor is local and often has low bias, but it is noisy. Many neighbors reduce variance but can introduce bias by including less comparable controls. Strict calipers improve credibility of matches but may change the target population.

Bias from Imperfect Matches

Let

$$m_0(x) = \mathbb{E}[Y_{0i} \mid X_i = x].$$

Suppose treated unit i is matched to control $m(i)$. Conditional on their covariates, the expected imputation error is approximately

$$\mathbb{E}[Y_{0,m(i)} - Y_{0i} \mid X_i, X_{m(i)}] = m_0(X_{m(i)}) - m_0(X_i).$$

If $m_0(\cdot)$ is smooth, a first-order Taylor approximation gives

$$m_0(X_{m(i)}) - m_0(X_i) \approx \nabla m_0(X_i)'(X_{m(i)} - X_i).$$

The bias is small when matched covariates are close or when the untreated outcome surface is flat. It is large when the match is far away in a direction that strongly predicts untreated outcomes.

Curse of Dimensionality

Suppose each covariate is split into five bins. With p covariates there are

$$5^p$$

cells. With $p = 2$, there are 25 cells. With $p = 5$, there are 3,125 cells. With $p = 10$, there are 9,765,625 cells. Most cells will be empty in ordinary datasets. Even nearest-neighbor matching suffers: in high dimensions, the nearest neighbor may still be far away.

This is why matching is attractive as an idea but difficult as a mechanical solution. Adding covariates can reduce omitted confounding, but it also makes good matches harder to find. The answer is not to throw in every available variable blindly. The answer is to choose pre-treatment confounders using economic reasoning, then diagnose whether the data contain credible matches.

Common Support and Design Diagnostics

Common support is not a decoration added after estimation. It is part of the research design. Ask:

- Do treated units have nearby controls?

- Do controls have nearby treated units if the target is the ATE?
- How many observations are dropped by calipers or trimming?
- Are results driven by a small set of heavily reused controls?
- Are covariates balanced after matching?

A matched sample can have fewer observations than the original sample. This is not automatically a problem. A smaller credible comparison is often better than a larger incredible comparison. But the population must be described honestly. If the method drops older high-income treated units because no controls are similar, the estimate is not the ATT for all treated units.

Propensity Scores

Definition 3 (Propensity score)
The propensity score is the conditional probability of treatment:

$$e(X_i) = \mathbb{P}(D_i = 1 \mid X_i).$$

Propensity scores compress the treatment-selection information in X_i into a scalar. Units with the same propensity score have the same predicted probability of treatment, even if their covariate profiles are not identical. This can make matching, stratification, trimming, and weighting more feasible.

Theorem 1 (Propensity-score theorem)
If

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i \mid X_i,$$

then

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i \mid e(X_i),$$

provided overlap holds. The propensity score is a balancing score.

Balancing Intuition

The key balancing statement is

$$D_i \perp\!\!\!\perp X_i \mid e(X_i).$$

Within a narrow propensity-score stratum, treated and untreated units should have similar distributions of the covariates used to estimate the score. This does not mean each treated unit has a perfect twin. It means the distribution of covariates should be balanced between treatment groups within score strata or after weighting.

Proof 1 (Why the score balances covariates)
Let $e(X) = \mathbb{P}(D = 1 \mid X)$. For any value p in the support of $e(X)$,

$$\mathbb{P}(D = 1 \mid X = x, e(X) = p) = \mathbb{P}(D = 1 \mid X = x) = e(x) = p.$$

Therefore, within the set of units with $e(X) = p$, the probability of treatment does not depend on the remaining details of X. This is the sense in which the score balances covariates.

Proof 2 (Why conditioning on the score identifies effects)
If conditional independence holds given X, then for $d \in \{0, 1\}$,

$$\mathbb{E}[Y_d \mid D = 1, e(X)] = \mathbb{E}\{\mathbb{E}[Y_d \mid D = 1, X] \mid D = 1, e(X)\}.$$

By conditional independence,

$$\mathbb{E}[Y_d \mid D = 1, X] = \mathbb{E}[Y_d \mid X].$$

By the balancing property, the distribution of X conditional on $D = 1, e(X)$ equals the distribution conditional on $D = 0, e(X)$. Therefore

$$\mathbb{E}[Y_d \mid D = 1, e(X)] = \mathbb{E}[Y_d \mid D = 0, e(X)].$$

Thus treated and untreated units with the same propensity score can be compared under the same selection-on-observables assumption.

> **Beginner trap.** A propensity-score model is not judged mainly by its classification accuracy or AUC. A model that predicts treatment almost perfectly may create terrible overlap. The goal is balance and common support, not a beautiful treatment-choice regression table.

Estimating Propensity Scores

In practice $e(X_i)$ is unknown, so we estimate it:

$$\widehat{e}(X_i) = \widehat{\mathbb{P}}(D_i = 1 \mid X_i).$$

A common starting point is logistic regression,

$$\log\left(\frac{e(X_i)}{1 - e(X_i)}\right) = X_i'\gamma.$$

Researchers may also use probit, series terms, interactions, random forests, or gradient boosting. More flexible models can help balance, but they can also overfit. In applied work, the score model should be treated as a design model: its job is to make treated and untreated units comparable on pre-treatment covariates.

Covariate choice. Include pre-treatment variables that predict both treatment and potential outcomes. Do not include variables caused by treatment. Be careful with pure instruments: variables that strongly predict treatment but not outcomes can worsen overlap and inflate weights.

Diagnostics After Estimating the Score

After estimating $\widehat{e}(X)$, inspect:

- histograms or density plots of scores by treatment status;
- minimum, median, and maximum score in each group;
- the fraction of observations near 0 or 1;
- standardized mean differences before and after adjustment;
- maximum weights and effective sample size;
- which units are dropped by common-support restrictions.

For a covariate X_j, the standardized mean difference is

$$\mathrm{SMD}_j = \frac{\bar{X}_{j,1} - \bar{X}_{j,0}}{s_j},$$

where s_j is a pooled or full-sample standard deviation. Weighted versions replace the group means by weighted means. Values close to zero indicate better balance. The threshold 0.10 is often used as a rule of thumb, but it is not a law of nature.

Trimming and Propensity-Score Matching

Trimming removes observations with weak overlap:

$$a \leq \widehat{e}(X_i) \leq b.$$

Common choices include fixed cutoffs such as $0.05 \leq \widehat{e}(X_i) \leq 0.95$, or the empirical common support. Define

$$\ell = \max\{\min_{D_i=1} \widehat{e}(X_i), \min_{D_i=0} \widehat{e}(X_i)\}, \qquad u = \min\{\max_{D_i=1} \widehat{e}(X_i), \max_{D_i=0} \widehat{e}(X_i)\}.$$

Then keep observations satisfying

$$\ell \leq \widehat{e}(X_i) \leq u.$$

Trimming is not a harmless numerical step. It changes the estimand to the population that remains after trimming.

Propensity-score matching uses the scalar distance

$$d(i, \ell) = |\widehat{e}(X_i) - \widehat{e}(X_\ell)|.$$

The advantage is simplicity. The danger is that two units with the same score can have different covariate profiles. Therefore balance must be checked on the original covariates, not only on the score.

Inverse Probability Weighting

Inverse probability weighting reweights observations by how rare their treatment status is for their covariates. For the ATE, the population moments are identified by

$$\mathbb{E}[Y_1] = \mathbb{E}\left[\frac{D_i Y_i}{e(X_i)}\right], \qquad \mathbb{E}[Y_0] = \mathbb{E}\left[\frac{(1-D_i)Y_i}{1-e(X_i)}\right].$$

Therefore

$$\text{ATE} = \mathbb{E}\left[\frac{D_i Y_i}{e(X_i)} - \frac{(1-D_i)Y_i}{1-e(X_i)}\right].$$

Proof 3 (IPW identity for Y_1)
Using iterated expectations,

$$\mathbb{E}\left[\frac{D_i Y_i}{e(X_i)}\right] = \mathbb{E}\left[\mathbb{E}\left(\frac{D_i Y_{1i}}{e(X_i)} \mid X_i\right)\right].$$

Under conditional independence,

$$\mathbb{E}[D_i Y_{1i} \mid X_i] = \mathbb{E}[D_i \mid X_i]\mathbb{E}[Y_{1i} \mid X_i] = e(X_i)\mathbb{E}[Y_{1i} \mid X_i].$$

Thus

$$\mathbb{E}\left[\frac{D_i Y_i}{e(X_i)}\right] = \mathbb{E}[\mathbb{E}(Y_{1i} \mid X_i)] = \mathbb{E}[Y_{1i}].$$

The proof for Y_0 is analogous.

A normalized, Hajek-style ATE estimator is

$$\widehat{\text{ATE}}_{IPW} = \frac{\sum_i D_i Y_i / \widehat{e}(X_i)}{\sum_i D_i / \widehat{e}(X_i)} - \frac{\sum_i (1-D_i) Y_i / [1-\widehat{e}(X_i)]}{\sum_i (1-D_i) / [1-\widehat{e}(X_i)]}.$$

The treated group is weighted to represent everyone. The untreated group is also weighted to represent everyone. The difference estimates the ATE under conditional independence and overlap.

IPW for the ATT

For the ATT, treated units already represent the treated population. Controls must be reweighted to look like treated units:

$$w_i^{ATT} = \begin{cases} 1, & D_i = 1, \\ \dfrac{e(X_i)}{1-e(X_i)}, & D_i = 0. \end{cases}$$

A normalized ATT estimator is

$$\widehat{\text{ATT}}_{IPW} = \frac{\sum_i D_i Y_i}{\sum_i D_i} - \frac{\sum_i (1 - D_i) w_i^{ATT} Y_i}{\sum_i (1 - D_i) w_i^{ATT}}.$$

Controls with high treatment probability receive larger weight because they look like the treated population.

Weight Diagnostics

Extreme propensity scores create extreme weights. For the ATE, a treated unit with $\widehat{e}(X_i)$ close to zero receives weight $1/\widehat{e}(X_i)$. A control unit with $\widehat{e}(X_i)$ close to one receives weight $1/[1 - \widehat{e}(X_i)]$. A few observations can dominate the estimate.

Report at least the maximum weight, selected percentiles, and effective sample size:

$$ESS = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

If 2,000 observations have an effective sample size of 60 after weighting, the design is fragile even if the regression output prints a small standard error.

The Companion Python Example

The script `lectures/code/lecture-6-matching-propensity.py` simulates an observational study with selection on observables. Treatment is more likely for some observed covariate profiles, treatment effects are heterogeneous, and the true ATE and ATT are known because the data are simulated.

Students should run

```
python lectures/code/lecture-6-matching-propensity.py
```

and compare:

- the naive treated-control gap;
- regression-adjusted OLS;
- one-nearest-neighbor matching for the ATT;
- propensity-score IPW for the ATE;
- propensity-score weights for the ATT;
- trimmed IPW estimates.

The useful lesson is not that one estimator always wins. The useful lesson is that the same data can produce different estimates because methods target different populations and rely on different design choices.

Common Mistakes

Mistakes to avoid.

- Matching on post-treatment variables.
- Reporting a propensity-score regression table instead of balance.
- Estimating an ATE when there is overlap only for the treated population.
- Keeping extreme weights because the preferred estimate depends on them.
- Checking balance only on the propensity score and not on covariates.
- Forgetting that trimming, calipers, and dropped observations change the target population.
- Calling matching causal without defending conditional independence.

Reporting a Matching or Propensity-Score Study

A credible report should state:

1. the causal estimand: ATE, ATT, ATU, or an overlap-population effect;
2. the identifying assumption and why it is plausible in the setting;
3. the list of covariates and why they are pre-treatment confounders;
4. the matching method or propensity-score model;
5. overlap diagnostics;
6. balance before and after adjustment;
7. number of observations dropped, matched, or reused;
8. weight diagnostics if weighting is used;
9. standard errors and how they account for the full procedure;
10. sensitivity of results to calipers, trimming, covariates, and target estimand.

The reader's main question is: after this design step, would I believe treated and comparison units are exchangeable with respect to potential outcomes?

Worked Problems

Example 3 (Exact matching ATT)
There are two cells. In cell A, treated mean outcome is 12 and control mean outcome is 9. In cell B, treated mean outcome is 20 and control mean outcome is 18. Among treated units, 70 percent are in cell A and 30 percent are in cell B. The ATT is

$$(12 - 9) \times 0.70 + (20 - 18) \times 0.30 = 2.1 + 0.6 = 2.7.$$

The raw treated-control gap would generally be different because it also uses the untreated distribution across cells.

Example 4 (ATT weights)
Suppose a control unit has $\widehat{e}(X) = 0.80$. Its ATT control weight is

$$\frac{0.80}{1 - 0.80} = 4.$$

This control receives high weight because its covariates make it look like a unit that was likely to be treated. If another control has $\widehat{e}(X) = 0.20$, its ATT weight is $0.20/0.80 = 0.25$. It contributes less to the treated counterfactual.

Example 5 (Effective sample size)
Consider weights $w = (1, 1, 1, 10)$. The nominal sample size is 4, but

$$ESS = \frac{(1 + 1 + 1 + 10)^2}{1^2 + 1^2 + 1^2 + 10^2} = \frac{169}{103} \approx 1.64.$$

The weighted estimate is effectively using far less information than the raw sample size suggests.

Checklist for Students

1. Write the potential-outcomes estimand before choosing a method.
2. State whether the target is ATE, ATT, ATU, or a trimmed/overlap population.
3. Use only pre-treatment covariates for the design.
4. Defend conditional independence using the institutional setting.
5. Check overlap before trusting estimates.

6. For matching, report the distance metric, replacement rule, number of neighbors, caliper, and dropped units.
7. For propensity scores, report balance and overlap, not only the score model.
8. For IPW, report weight percentiles, maximum weights, and effective sample size.
9. Check standardized mean differences before and after adjustment.
10. Explain how trimming or matching changes the population being studied.

Practice Questions

1. In your own words, explain why the raw treated-control difference can be biased in an observational study.
2. Derive the exact-matching formula for the ATT using iterated expectations and conditional independence.
3. Give an example of a post-treatment variable and explain why matching on it can be harmful.
4. Why does exact matching become infeasible when many covariates are continuous?
5. In nearest-neighbor matching, what is the tradeoff between one neighbor and many neighbors?
6. What does overlap mean for the ATT? What does it mean for the ATE?
7. State the propensity-score theorem and explain why it is useful.
8. Why is high treatment-prediction accuracy not the main goal of a propensity-score model?
9. Derive the IPW identity $\mathbb{E}[D_i Y_i / e(X_i)] = \mathbb{E}[Y_{1i}]$.
10. A control unit has $\widehat{e}(X) = 0.90$. What is its ATE control weight? What is its ATT control weight? What does this imply about overlap?
11. Run the companion Python script. Which estimator targets the ATT? Which targets the ATE? How do trimming and weights change the interpretation?
12. Suppose a paper reports propensity-score matching but no balance table. What should you ask the author to provide?